PROJECT-BASED LEARNING

PHASE-II EVALUATION

CAMPUS SKILL SWAP NETWORK

Team ID: DSCPP-III-2026-T178

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T178
Semester: 3rd
Section: C, D, F, G
Domain: *Peer Learning / C++ Application*

Team Members

1. *Mayank Rawat – 2510010445*
2. *Pranav Uniyal – 2510010601*
3. *Vaibhavi Saklani – 2510011544*
4. *Muskan – 2510010799*

Mentor

Dr. Narayan Chaturvedi

Interactions completed: _____

Phase-I Recap & Feedback Incorporated

Phase-I Status

In Phase-I, we studied existing skill-sharing systems, defined the key requirements, designed the system architecture, developed a basic skill-matching module, and designed the student and skill data storage.

Feedback Received

- Project scope was too broad for the available time.
- Objectives were not clearly measurable.
- Course concepts were listed without clear implementation.

Changes Incorporated

- Focused on core modules: profiles, matching, exchange requests and scheduling. Chat deferred to Phase-III.
- Defined four measurable objectives.
- Listed only the DSA and OOP concepts actually implemented.

Problem Statement & Finalized Objectives

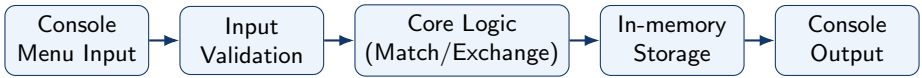
Problem Statement

Students have different skills but struggle to find suitable classmates to learn from and teach. Our system connects students based on complementary skills and interests, making peer learning easier and more collaborative.

Finalized Objectives

1. Register and authenticate students with skills to teach/learn (with levels) and preferences.
2. Match students on complementary skills and rank them by compatibility.
3. Manage the full exchange lifecycle: request, accept/reject, schedule and complete.
4. Collect ratings and feedback, and provide admin monitoring with system statistics.

Updated System Architecture / Workflow



Major Components

- Profile & Preference Manager
- Skill Manager
- Match Engine (matching, ranking, search)
- Exchange & Scheduling Workflow
- Feedback & Notifications
- Admin Panel

Changes Since Phase-I

- Added Skill Evolution Map (learning path between skills)
- Added compatibility ranking and multi-filter search
- Added notifications and admin statistics
- Chat system deferred to Phase-III

Module Design & Functional Requirements

Major Modules

1. Profile & Preference Manager
2. Skill Manager (teach / learn lists)
3. Match Engine
4. Exchange & Scheduling Workflow
5. Feedback, Notifications & Admin

Functional Requirements

- Register / login with unique Student ID
- Add / remove skills with level (Beginner to Advanced)
- Find and rank matches; search by 7 filters
- Request, accept / reject / cancel, then schedule
- Rate (1–5) and give feedback after exchange

Module Dependency

Match Engine reads data from the Profile and Skill modules; the Exchange Workflow uses Match Engine results; Feedback and Notifications are triggered by exchange events; Admin reads all modules.

Technology Stack & Development Methodology

Technology Stack

- Programming: *C++ (console application)*
- Storage: *In-memory (STL containers)*
- Framework: *None (C++ STL)*
- Development Tool: *Visual Studio Code*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C++	*Vectors (dynamic arrays), graph traversal for learning path, sorting for compatibility ranking*	*Match Engine, Skill Evolution Map*
OOPs with C++	*Classes and encapsulation, STL (vector, string)*	*Profile Manager, Exchange Workflow*
Operating Systems	*Not implemented in this phase*	–
DBMS	*Planned: persistent file / database storage*	*Phase-III*

Development & Implementation Progress

Module	Status	Work Completed
Profile & Login	Completed	Registration with ID check, login, profile edit, preferences
Skill Manager	Completed	Teach / learn lists with levels, add, remove, display
Match Engine	Completed	Skill matches, compatibility ranking, 7-filter search, Skill Evolution Map
Exchange Workflow	Completed	Send / accept / reject / cancel requests, scheduling, skill history
Feedback, Notif. & Admin	Completed	1–5 rating with feedback, notifications, admin statistics
Data Persistence	Pending	Currently in-memory; file / DB storage planned
Chat System	Pending	Planned for Phase-III

Overall Project Completion: *75%*

Working Prototype / Demonstration

Implemented Features

- Student registration, login and profile management
- Skill matching with compatibility ranking
- Exchange requests with scheduling
- Ratings, notifications and admin panel

Prototype / Output

Insert console screenshots here (main menu, match results, exchange request)

Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Duplicate Student ID	Registration rejected	"Student ID already exists" shown	Pass
Invalid year (outside 1–4)	Re-prompt for valid year	Re-prompted until valid	Pass
Request to self	Request blocked	"Cannot send request to yourself"	Pass
Schedule non-accepted request	Scheduling blocked	"Only accepted requests can be scheduled"	Pass
Rating outside 1–5	Re-prompt for valid rating	Re-prompted until valid	Pass
Invalid admin ID	Access denied	"Invalid Admin ID" shown	Pass

GitHub Repository & Development Activity

Repository Information

- GitHub Link:
  `https://github.com/mayankrawat5/Campus-skill-swap-`
- README.md: *Updated*
- Contributors: *All 4 members added*
- Repository Status: *Updated*

Development Activity

- *Regular commits maintained*
- *Individual contribution visible*
- *Latest implementation uploaded*
- *Documentation maintained*

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Mayank Rawat	*Lead*	*Matching and Connection module,Match Ranking and graph - based skill connections(BFS/DFS)*	*XX*
Pranav Uniyal	*Developer*	*Skill Management and skill search module , including teach or learn skills and search filters*	*XX*
Vaibhavi Saklani	*DB / Testing*	*User interface and integration of project modules*	*XX*
Muskan	*Developer/Docs*	*User Registration , login , profile creation and Profile management module*	*XX*

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- Validating user input across many menu options
- Keeping request status consistent across the exchange lifecycle
- Finding a learning path between related skills

Solutions / Learning

- Re-prompt loops with clear error messages
- Modular design with separate classes per feature

Phase-III Roadmap

1. Add persistent storage (file / database)
2. Add chat system between matched students
3. Complete system integration and optimization
4. Final testing and validation
5. Final documentation and report

Expected Outcomes & References

Expected Final Outcomes

- Complete working skill swap system with persistent data
- Reliable matching verified through test cases
- Practical use for peer learning on campus
- Scalable to multiple colleges with a database backend

References

1. D.S. Malik, *Using C++*, Cengage Learning
2. T.H. Cormen et al., *Introduction to Algorithms*, MIT Press
3. D. Gale and L.S. Shapley, "College Admissions and the Stability of Marriage", *American Mathematical Monthly*, 1962
4. C++ Reference: `cppreference.com`

Thank You

Questions & Suggestions